CriminalID — Face Recognition System

Streamlit Web App | Transfer Learning | 97.85% Accuracy

Project Structure

```
your_project_folder/
├── CriminalID.py                        ← Streamlit app
├── criminal_recognition_model.keras     ← Trained model (Transfer Learning)
├── criminals_info.csv                   ← Criminal profile database
├── class_names.txt                      ← 13 class names (alphabetical)
├── requirements.txt                     ← Python dependencies
├── Dataset_Creation.ipynb               ← Dataset prep notebook
├── Criminal_Face_Recognition_Using_Transfer_Learning.ipynb
└── criminal_dataset/                    ← Training images (13 classes)
```

Model Performance

Metric	Value
Overall Validation Accuracy	**97.85%**
Macro Avg Precision / Recall / F1	0.98 / 0.98 / 0.98
Total Training Samples	17,680 (13 classes × 1360 each)
Training Strategy	Frozen base → Fine-tuning (epoch 13)

Recognized Individuals — 13 Classes

Aniket Kakad, Atharva Ayachit, Chandrakant Kshirsagar, Dhananjay Patil, Ishika Mehre, Japneet Singh, Jay Kumar, Jyoti Ayachit, Kajal Yerone, Sakshi Lahekar, Sayali Chidrawar, Shreyas Jadhav, Tushar Patil

Setup & Run

1. Install Dependencies

```
pip install -r requirements.txt
```

Or manually:

```
pip install streamlit tensorflow opencv-python-headless pillow pandas numpy
```

2. Prepare Files

- Train the model using the notebook → saves as criminal_recognition_model.keras
- Ensure criminals_info.csv exists with the correct columns
- class_names.txt contains 13 names in alphabetical order

3. Run the App

```
streamlit run CriminalID.py
```

Then open http://localhost:8501 in your browser.

How to Use

Upload Image Tab

1. Upload any photo (jpg/png/webp)
2. Click RUN IDENTIFICATION
3. If a face is found → model classifies it → result shown side by side

Webcam Tab

4. Click the camera widget → allow browser camera access
5. Take a photo → auto-runs detection immediately

Database Tab

View all criminal records. Search and filter by name or crime.

Sidebar Settings

Setting	Description
Model Path	Path to your .keras model file
Criminal Info CSV	Path to criminal profiles CSV
Confidence Threshold	Min % confidence to make a positive ID (default 80%)
Class Names File	Optional .txt with class names in order

criminals_info.csv Format

```
name,crime,status,age,nationality,last_seen,description
Aniket_Kakad,Theft,Wanted,22,Indian,Pune 2024,Considered dangerous
```

Note: name must exactly match folder names. Status values: Wanted / Arrested / Imprisoned

Deployment

This app is deployed on Streamlit Community Cloud. Visit the live app at:

```
https://YOUR_APP_NAME.streamlit.app
```

Links

- GitHub Repository: https://github.com/YOUR_USERNAME/CriminalID-Face-Recognition
- Live App: https://YOUR_APP_NAME.streamlit.app